Labour Market Integration of Immigrants in Canada Post-Pandemic (2021)

A Predictive and Descriptive Analysis

(Final Report)

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Research Questions & Contribution

This report addresses the following research questions:

1. To what extent can employment outcomes in Canada be predicted using observable demographic, education-related, and geographic characteristics?
2. Do age, education, gender, and geography help explain differences in employment outcomes across immigrant and non-immigrant groups?
3. Which modelling approach provides the strongest balance of effectiveness, efficiency, and stability for predicting employment status using the available features?
4. What do the remaining employment gaps suggest about structural or institutional barriers not captured in the dataset?

Main Contribution of the Study

This study contributes to the literature in three key ways:

First, it combines predictive modelling with descriptive labour market analysis. Rather than treating model performance as the sole outcome, the study uses predictive results to evaluate how well observable variables explain employment status, and complements this with subgroup comparisons across age, education, gender, and geography categories.

Second, the study is situated in a post-pandemic labour market context. While the analysis is based on cross-sectional data from the 2021 Census of Population, it is interpreted in light of recent labour market shifts following COVID-19, including changes in employment patterns and immigrant integration outcomes (Hou, 2024).

Third, the study provides a structured model evaluation framework that compares classifiers based not only on effectiveness, but also on efficiency and stability. This allows for a more comprehensive assessment of model performance beyond predictive accuracy alone.

The core substantive contribution of the study is the finding that observable demographic, educational, and geographic characteristics are important correlates of employment outcomes, but they do not fully explain the persistent differences in employment rates between immigrant and non-immigrant groups. This suggests that additional structural, institutional, and regional factors play a meaningful role in shaping labour market integration.

Introduction

Labour market integration remains one of the most important dimensions of immigrant settlement in Canada. Immigration plays a central role in population growth, labour force sustainability, and long-term economic development. However, the transition into stable and equitable employment is not uniform across population groups. Despite Canada's points-based immigration system, which emphasizes education and labour market potential, differences in employment outcomes between immigrants and non-immigrants have historically persisted (OECD, 2019; Kaushik & Drolet, 2018).

Recent evidence suggests that immigrant labour market outcomes have improved since the mid-2010s, with employment rates rising and the overall gap between immigrants and the Canadian-born population narrowing (Hou, 2024). However, improvement does not necessarily imply full convergence. Differences in employment outcomes may persist even when immigrants possess similar levels of formal education or fall within comparable age groups as non-immigrants.

This study examines whether observable demographic, education-related, and geographic characteristics, specifically age, education, gender, and geography, can explain employment outcomes and whether differences between immigrant and non-immigrant groups remain after accounting for these factors. The analysis is based on aggregated data from the 2021 Census of Population, which provides detailed cross-tabulations of labour market outcomes across demographic, educational, and geographic categories.

The study adopts a mixed analytical approach. The first component uses machine learning models to predict employment status from selected features. The second component uses descriptive subgroup analysis to compare employment rates across immigrant status, age groups, education levels, gender, and geography. Together, these approaches allow the study to evaluate both predictive performance and substantive labour market patterns.

The report is structured as follows. It first reviews relevant literature on immigrant labour market integration, then outlines the data, preprocessing steps, and modelling approach. It next evaluates model performance based on effectiveness, efficiency, and stability. The findings are then interpreted in the context of employment differences across demographic, educational, and geographic groups. The final sections discuss limitations, ethical considerations, and directions for future research.

Literature Review & Integration

Labour Market Integration as a Multi-Dimensional Process

Labour market integration is widely understood as a multi-dimensional process that extends beyond formal human capital. While Canada's immigration system emphasizes education, skills, and labour market potential, research consistently shows that these characteristics do not automatically translate into equivalent employment outcomes.

The OECD (2019) highlights that immigrant qualifications and selection criteria do not guarantee successful labour market integration, even in a points-based system. Similarly, Kaushik and Drolet (2018) find that skilled immigrants in Canada continue to face barriers despite high levels of formal education. These findings help explain why strong educational attainment among immigrants may coexist with weaker employment outcomes or lower returns to education.

More recent Canadian evidence suggests that immigrant labour market outcomes have improved over time. Hou (2024) shows that employment rates among recent immigrants increased significantly between 2010 and 2023, narrowing the gap relative to the Canadian-born population. However, these improvements do not imply full convergence, as differences may persist within specific demographic or educational subgroups.

Credential Recognition, Work Experience, and Skill Transferability

One of the most important explanations for persistent employment differences is the incomplete transferability of foreign-acquired education and work experience. The Evaluation of the Foreign Credential Recognition Program (ESDC, 2025) identifies several key barriers faced by internationally trained individuals, including the time and cost of credential recognition, limited access to Canadian work experience, and challenges navigating professional licensing systems.

Kaushik and Drolet (2018) similarly emphasize institutional and structural barriers, including limited professional networks and difficulties accessing appropriate employment opportunities. These findings are particularly relevant to the present study. If employment differences persist even after accounting for age and education, it suggests that the remaining gap may be driven by institutional factors rather than differences in formal qualifications alone.

Language and Employment Outcomes

Language proficiency is another important factor linking immigrant characteristics to labour market outcomes. Xu and Hou (2023) find that official language proficiency is associated with improved labour market outcomes, particularly in terms of earnings. While its effect on initial employment may be more limited, language plays a significant role in long-term labour market integration and career progression.

This distinction is important for the current study. Even when age and education are observed, employment outcomes may still reflect underlying differences in language ability, social capital, and labour market signalling mechanisms.

Integrating the Literature with This Study

The present study builds on this literature by testing whether observable demographic, education-related, and geographic characteristics can predict employment outcomes, help explain differences between immigrant and non-immigrant groups, and point to structural or institutional factors that remain outside the dataset.

The findings provide only partial support for a human capital explanation. While employment rates vary across age, education, gender, and geographic categories, differences between immigrant and non-immigrant groups persist within comparable subgroups. This pattern is more consistent with literature emphasizing institutional barriers, skill transferability, regional labour market conditions, and labour market frictions than with a model in which formal education and demographic characteristics fully determine employment outcomes (Kaushik & Drolet, 2018; OECD, 2019; Xu & Hou, 2023; ESDC, 2025).

In this sense, the study reinforces the view that labour market integration is not solely an individual-level outcome, but a process shaped by broader structural and institutional factors.

Methodology & Study Design

Data Source

This study uses aggregated data from **Statistics Canada Table 98-10-0442-01**, derived from the **2021 Census of Population**. The dataset provides cross-tabulated counts of labour force status by demographic and educational characteristics, including age group, gender, immigrant status, level of education, and field of study.

Unlike microdata sources such as the Labour Force Survey (LFS), this dataset is structured as an aggregated table. Each row represents a grouped combination of characteristics rather than an individual observation. As a result, the analysis is conducted at the group level, which has important implications for both modelling and interpretation.

The dataset contains a large number of observations due to the combination of multiple categorical variables across demographic and educational dimensions. This structure required preprocessing to isolate meaningful analytical units and remove non-informative aggregate categories.

Study Design

The project adopts a mixed-method approach:

1. A predictive modelling component, which frames employment status as a binary classification problem using aggregated observations.
2. A descriptive analytical component, which compares employment rates between immigrant and non-immigrant groups overall and within age, education, gender, and geography categories.

This design allows the study to evaluate both the predictive power of observable variables and the persistence of group-level employment differences.

Data Cleaning and Preprocessing

Several preprocessing steps were applied to prepare the dataset for analysis:

- Rows representing aggregate totals or non-specific categories (e.g., “Total”) were removed.

- The dataset was restricted to observations classified as either *Employed* or *Unemployed*, excluding individuals outside the labour force.
- A binary target variable was constructed:
    - 1 = Employed
    - 0 = Unemployed
- Missing values were handled within the preprocessing pipeline.
- Categorical variables were transformed using one-hot encoding.
- Numeric variables were standardized where applicable.

These steps were necessary to convert the aggregated dataset into a format suitable for classification modelling and subgroup analysis.

Feature Selection

The predictive models used the following features:

- Geography
- Gender
- Age group
- Highest certificate, diploma, or degree
- Location of study
- Major field of study

These variables were selected because they represent key demographic, education-related, and geographic characteristics that are plausibly associated with employment outcomes. They also align directly with the study's research questions regarding whether observable characteristics predict employment status, whether subgroup differences persist, and what remaining gaps suggest about structural or institutional barriers.

Model Selection

Two models were retained for the final analysis:

1. **Logistic Regression**
   This model serves as a transparent baseline and allows for interpretation of feature relationships through model coefficients.

2. **Random Forest**
   This model captures nonlinear relationships and interactions between variables and provides feature importance measures for comparison.

Gradient Boosting was initially explored but was excluded from the final workflow to maintain computational efficiency while still enabling meaningful model comparison.

Training, Testing, and Sampling Strategy

The dataset was divided into training and testing subsets using an 80/20 split with stratification on the target variable to preserve class balance.

Due to the size and structure of the dataset, stratified sampling was used to improve computational efficiency:

- Logistic Regression: 300,000 training rows
- Random Forest: 100,000 training rows

This approach is appropriate given the study's objective of comparative model evaluation rather than production-level optimization. The sampling strategy ensures reproducibility while maintaining sufficient data representation across categories.

Evaluation Metrics

Model performance was evaluated across three dimensions:

1. **Effectiveness**
   Measured using: Accuracy, Precision, Recall, F1 Score.
   The F1 score is emphasized as it balances precision and recall, providing a more informative measure when classification performance is weak or imbalanced.
2. **Efficiency**
   Measured using: Training time, Prediction time.
   These metrics assess whether increased model complexity leads to meaningful improvements in performance
3. **Stability**
   Stability was evaluated using 3-fold cross-validation on a stratified sample of 30,000 rows. This approach allows for assessment of model robustness across different data splits while maintaining computational feasibility.

Model Evaluation

Effectiveness

Model	Training Rows	Accuracy	Precision	Recall	F1 Score
Logistic Regression	300,000	0.4995	0.4995	0.5175	0.5083
Random Forest	100,000	0.4960	0.4964	0.5496	0.5216

Table 1: Model Performance Table

As shown in **Table 1** and illustrated in **Figure 1**, both models demonstrate weak predictive performance. Accuracy values remain close to 0.50, indicating that neither model performs substantially better than random classification. The F1 scores are also only marginally above 0.50, suggesting limited ability to balance precision and recall effectively.

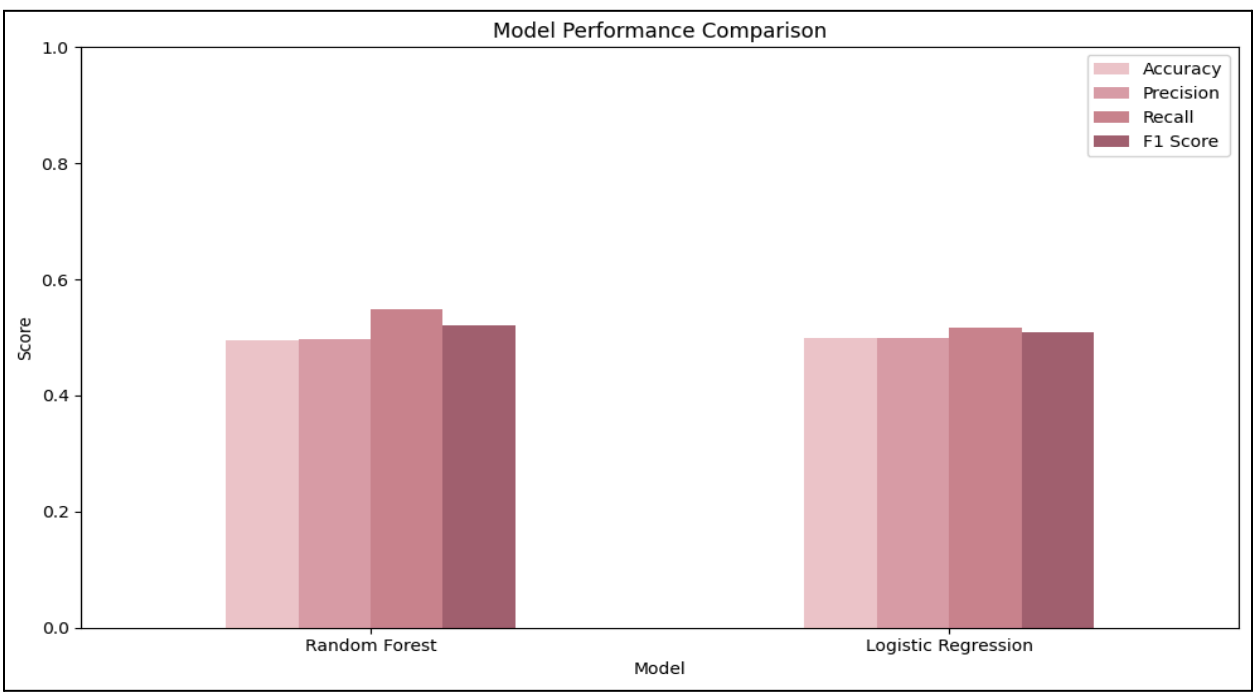

Figure 1: Model Performance Comparison

Accuracy and F1 Score Interpretation

Accuracy measures the share of predictions that were correct, while the F1 score summarizes the balance between precision and recall. In this study, Logistic Regression achieved an accuracy of 0.4995 and an F1 score of 0.5083, while Random Forest achieved an accuracy of 0.4960 and an F1 score of 0.5216. These values are close to 0.50, meaning that the models performed only slightly better than chance and should not be interpreted as strong predictive tools.

Among the two models, Random Forest achieves the highest F1 score (0.522), while Logistic Regression follows closely (0.508). However, as reflected in both the tabular results and the visual comparison in Figure 1, the difference between the models is minimal and does not represent a meaningful improvement in predictive capability.

This result is substantively important. The consistently weak performance across both models suggests that the selected demographic, education-related, and geographic features do not strongly determine employment outcomes within the dataset. In other words, while these variables are associated with employment, they are insufficient for accurate prediction on their own.

Efficiency

Model	Train Time (sec)	Test Time (sec)
Logistic Regression	0.60	0.44
Random Forest	0.48	0.61

Table 2: Efficiency Table

As presented in **Table 2** and visualized in **Figure 2**, both models demonstrate high computational efficiency under the sampled workflow. Logistic Regression has the fastest prediction time, while Random Forest shows slightly faster training time.

However, the differences are small and, as seen in Figure 2, do not meaningfully affect overall model comparison. Given the weak predictive performance observed in the effectiveness analysis, efficiency is not the deciding factor in model selection. Instead, the results suggest that increased model complexity does not lead to substantial gains in predictive accuracy within this feature set.

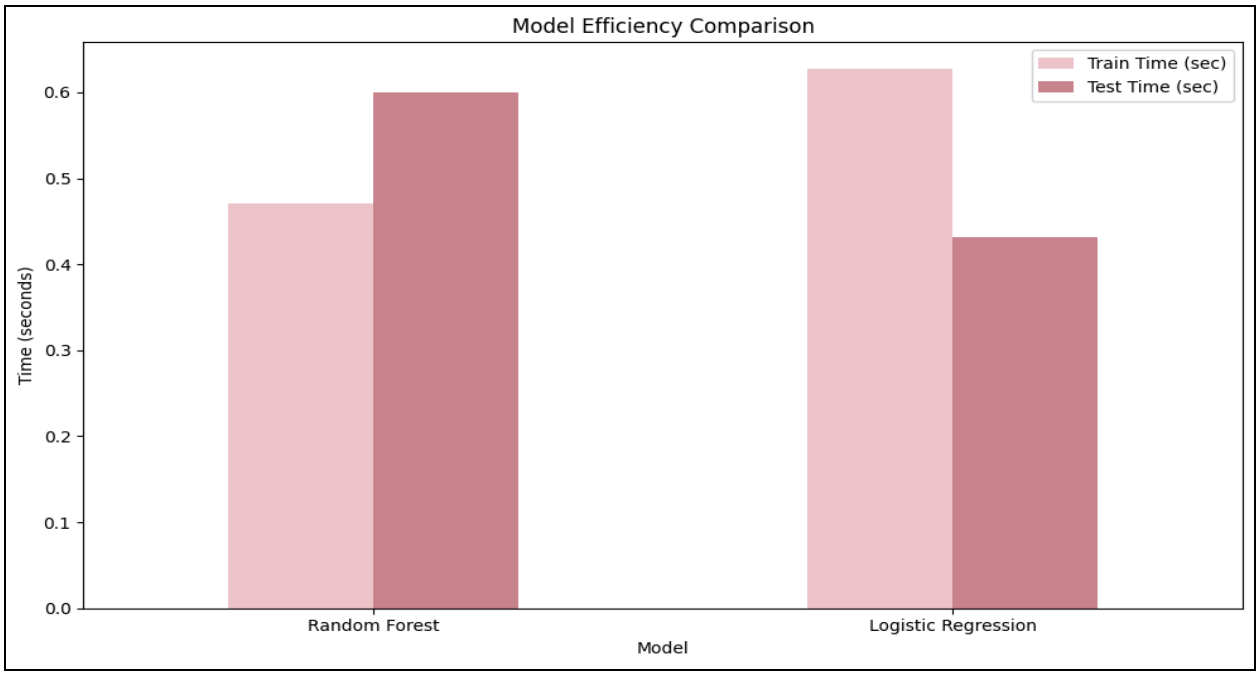

Figure 2: Model Efficiency Comparison

Stability

Model	CV Rows	Mean F1	Standard Deviation
Logistic Regression	30,000	0.4970	0.0034
Random Forest	30,000	0.4990	0.0158

Table 3: Cross-Validation Table

The cross-validation results shown in **Table 3** and summarized in **Figure 3** reinforce the findings from the test set. Both models produce mean F1 scores close to 0.50, indicating consistently weak predictive performance across different data splits.

Logistic Regression demonstrates lower variability across folds, suggesting more stable but still limited performance. Random Forest shows slightly higher average F1 scores but greater variability, as reflected in Figure 3, indicating less consistency across validation runs.

Overall, neither model demonstrates strong predictive stability in a substantive sense, as performance remains close to chance across all evaluation scenarios.

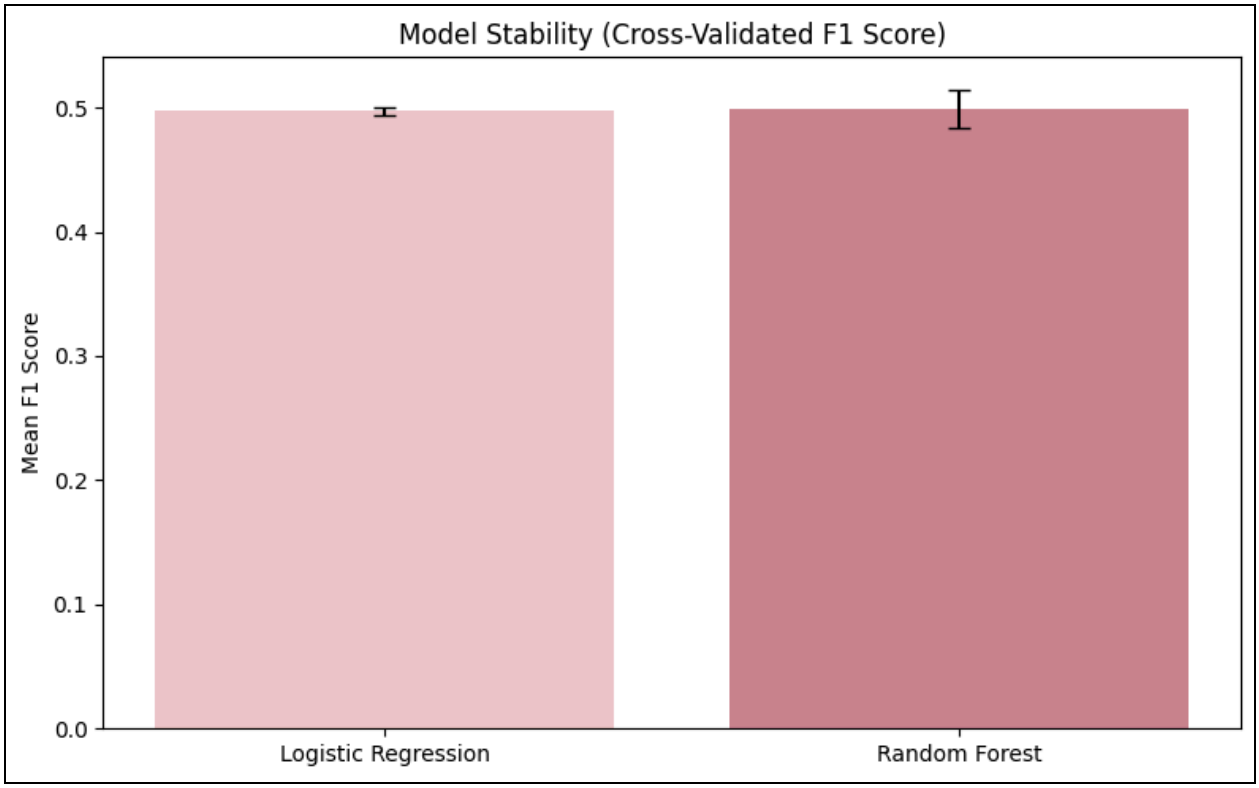

Figure 3: Cross-validated F1 comparison

Model Selection

The final model choice depends on the evaluation criteria emphasized:

- For interpretability and consistency: Logistic Regression is preferable
- For marginally higher F1 performance: Random Forest performs slightly better

However, when considering effectiveness, efficiency, and stability together, the most important conclusion is that neither model achieves strong predictive performance.

This conclusion is supported consistently across Tables 1–3 and Figures 1–3. Rather than indicating a limitation of the modelling approach alone, these results reflect the structure of the dataset and the scope of the selected features. Because the analysis is based on aggregated observations and a limited set of demographic, education-related, and geographic variables, the models are constrained in their ability to capture the full set of factors influencing employment outcomes.

The inclusion of both a linear model (Logistic Regression) and a nonlinear model (Random Forest) was intended to test whether more flexible modelling approaches would yield

meaningful improvements in performance. The results show that allowing for nonlinear relationships and interactions does not substantially improve predictive accuracy, suggesting that the limitation lies in the available features rather than in model complexity.

More complex models, such as Gradient Boosting, were considered but not retained in the final workflow. Given the minimal performance differences observed between the selected models, additional complexity would likely increase computational cost without providing meaningful gains in predictive performance.

Overall, the model comparison directly answers the predictive research question: observable demographic, education-related, and geographic variables do not strongly predict employment outcomes. This supports the interpretation that unobserved structural, institutional, and regional factors play a significant role in shaping labour market integration.

Findings & Interpretation

Overall Employment Comparison

The descriptive analysis shows that non-immigrants have a slightly higher overall employment rate than immigrants:

- Non-immigrant employment rate: **~92.8%**
- Immigrant employment rate: **~90.6%**

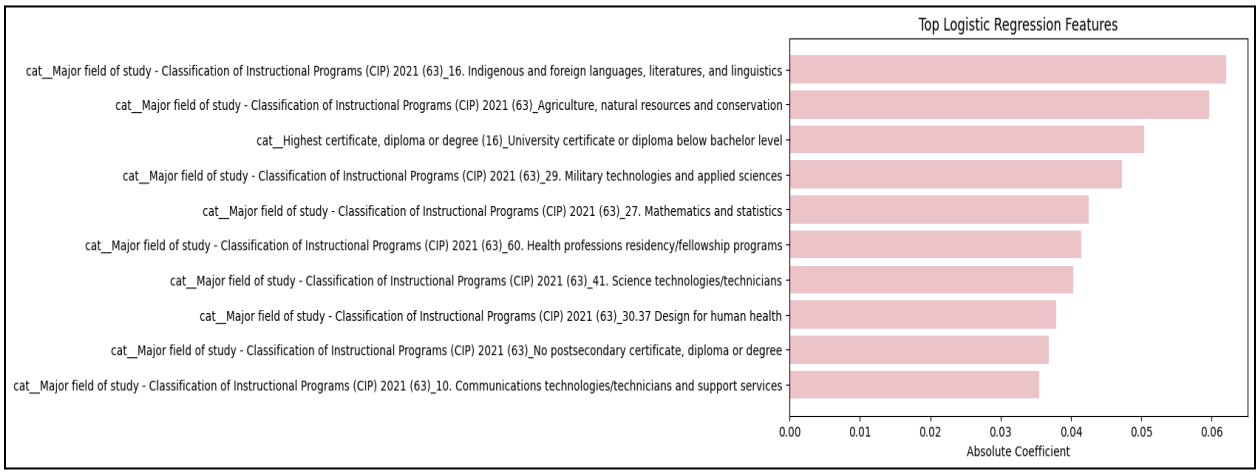

Figure 4: Top Logistic Regression features

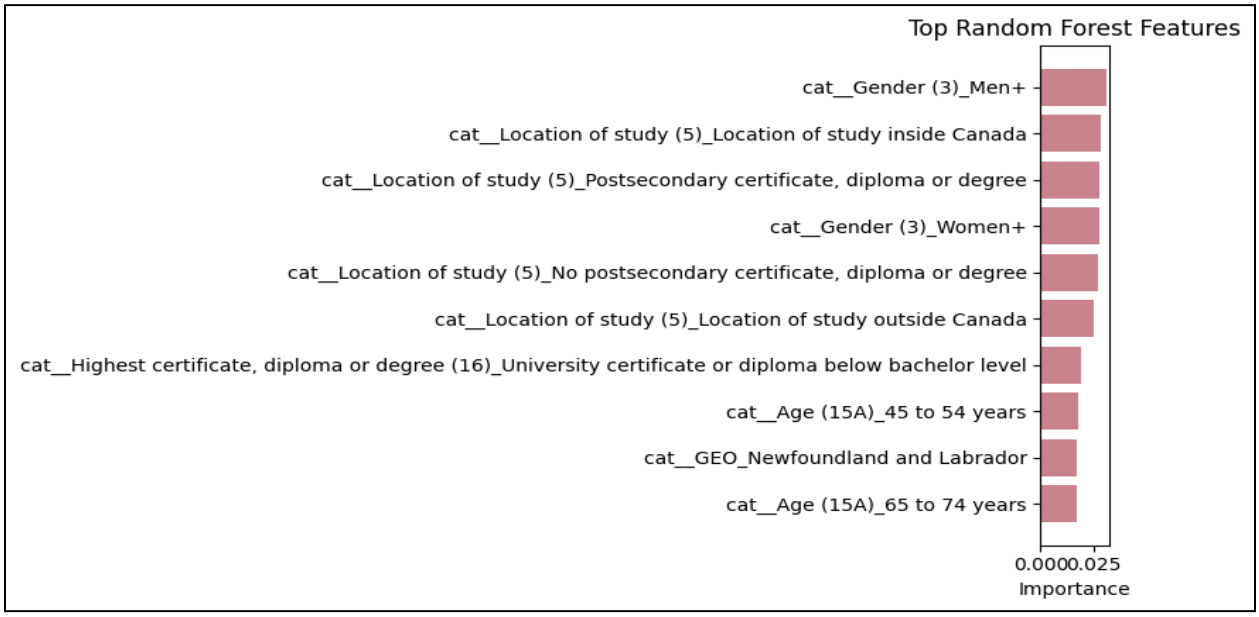

Figure 5: Top Random Forest features

Although the difference is modest, it is consistent. This establishes an important baseline pattern for the analysis. The key question is whether this gap disappears when comparing groups with similar levels of education, age, gender, and geography.

Feature importance results from both models (Figures 4 and 5) indicate that variables such as field of study, location of study, and age contribute to prediction. However, given the overall weak model performance, these results should be interpreted cautiously and used primarily to support broader descriptive findings rather than as definitive explanations.

Does Education Matter?

Education is clearly associated with employment outcomes, but it does not fully explain the difference between immigrant and non-immigrant groups.

As shown in **Figure 6**, employment rates are generally higher for individuals with stronger postsecondary qualifications across both groups. This aligns with established labour market expectations and with Canadian immigration policy, which prioritizes education and skill-based selection (OECD, 2019).

However, as illustrated in **Figure 7**, non-immigrants maintain slightly higher employment rates across most education categories. While the differences are not large, they are persistent rather than disappearing. This suggests that education alone is not sufficient to eliminate the employment gap.

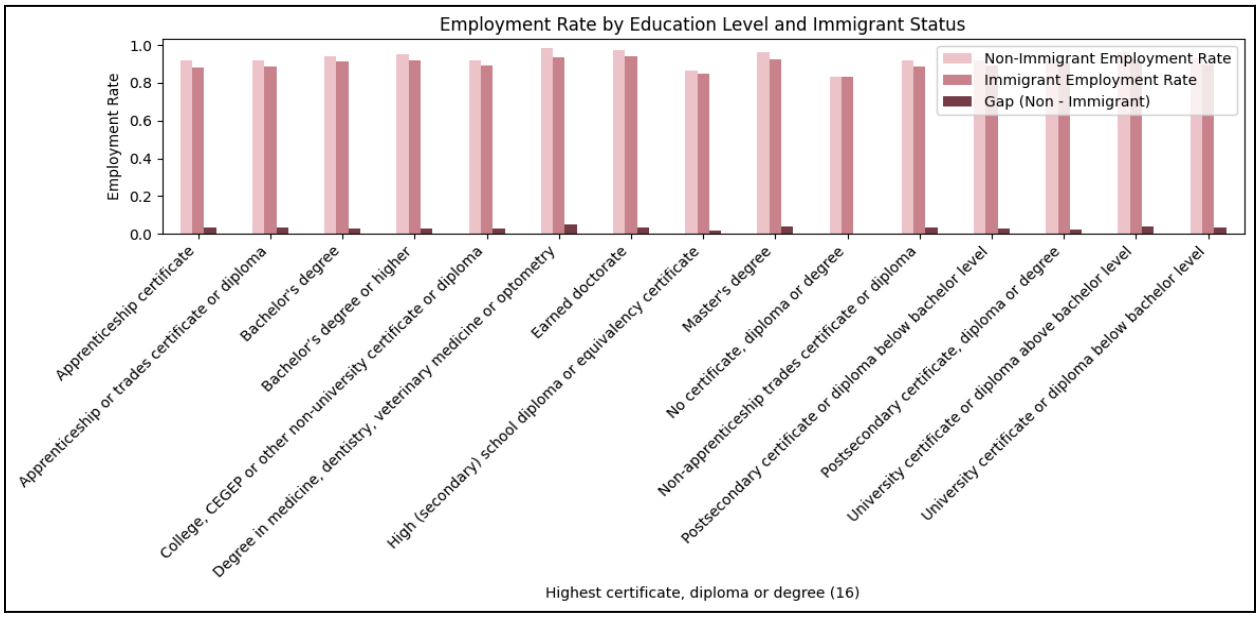

Figure 6: Employment rate by education level and immigrant status

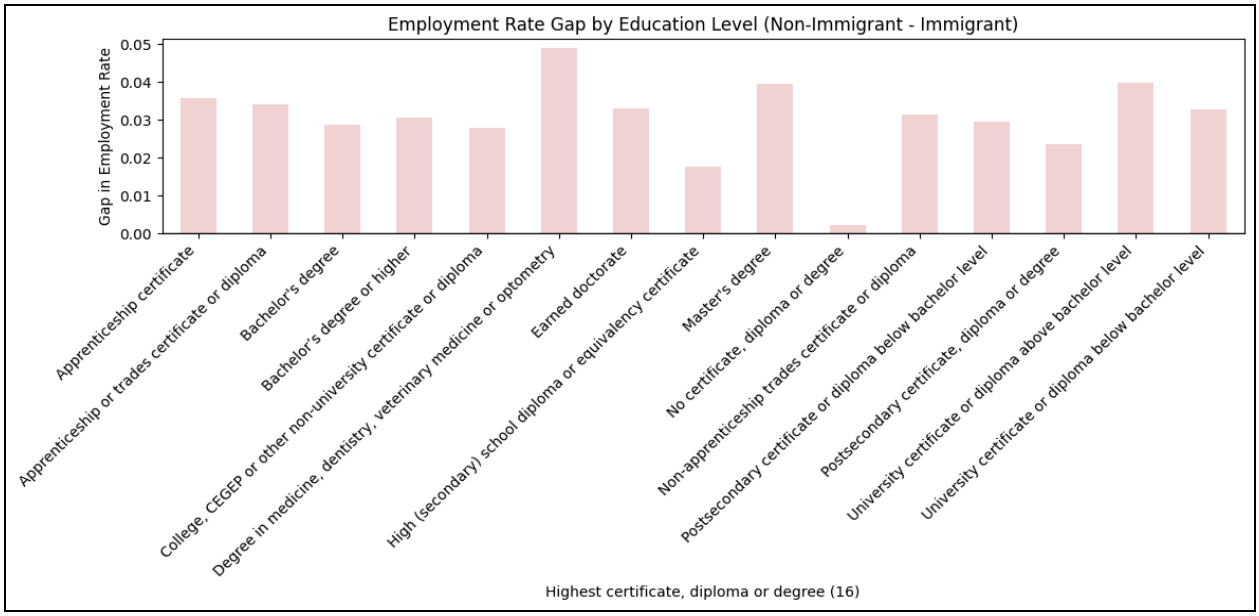

Figure 7: Employment rate gap by education level

This finding is consistent with prior research. The literature shows that the value of education depends on how it is recognized and utilized within the labour market. Factors such as location of study, credential recognition, and occupational requirements influence whether education translates into employment opportunities (Schimmele & Hou, 2024; ESDC, 2025).

In this context, the result can be summarized as follows:

- Education matters because employment rates vary across educational categories
- Education is not sufficient because group differences persist within those categories

Does Age Matter?

Age is also an important factor in employment outcomes, but it does not fully explain the observed group differences.

Employment rates vary across age groups, with lower attachment typically observed among younger individuals and those approaching retirement age. This pattern is consistent with expected labour market dynamics.

As shown in **Figure 8**, employment rates differ across age groups for both immigrants and non-immigrants. However, **Figure 9** shows that the gap between the two groups does not disappear within these categories. The differences tend to be more pronounced among younger age groups and smaller in some older categories, but they remain present overall.

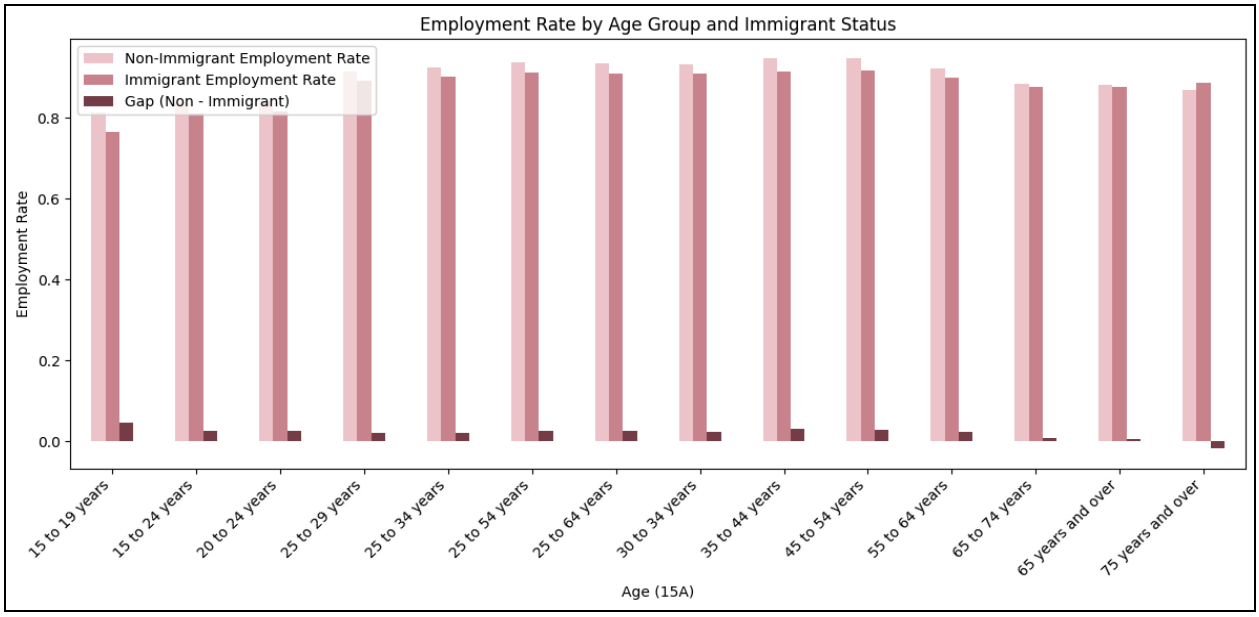

Figure 8: Employment rate by age group and immigrant status

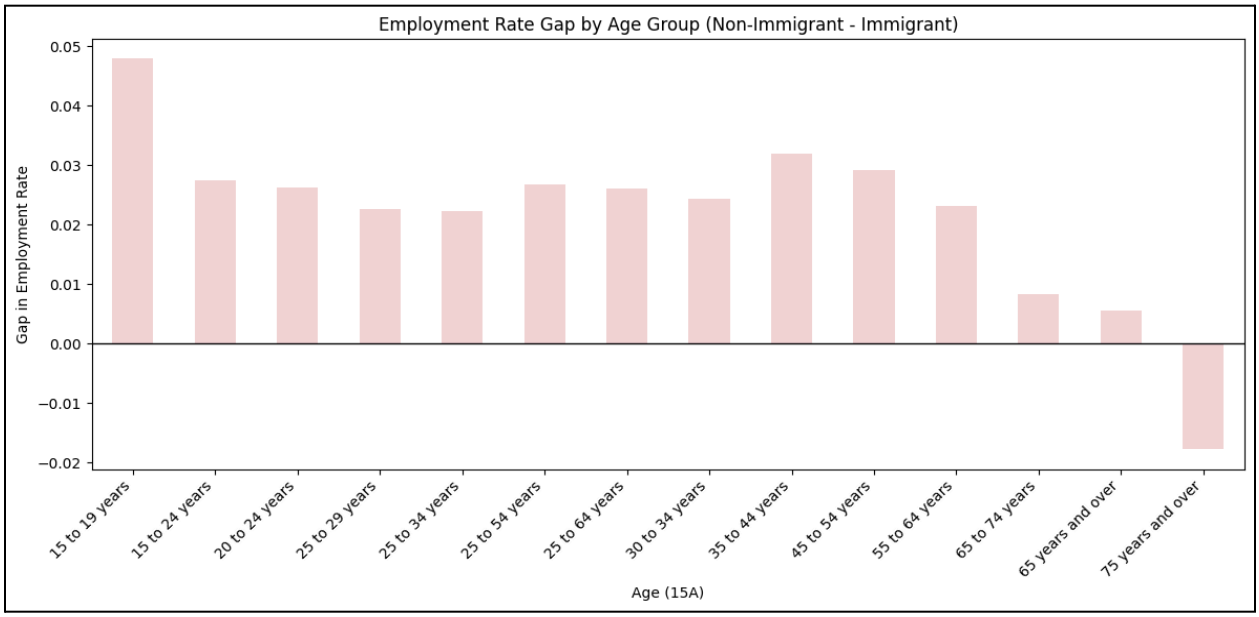

Figure 9: Employment rate gap by age group

This suggests that age plays a role in shaping employment outcomes but does not fully account for the persistent differences between groups.

One interpretation is that labour market entry may be particularly challenging for immigrants, especially younger individuals or recent entrants. This aligns with prior research highlighting barriers related to work experience, professional networks, and labour market entry pathways (Kaushik & Drolet, 2018; OECD, 2019).

Does Gender Matter?

Gender is also associated with employment outcomes, but the pattern differs across gender categories. As shown in Figure 10, employment rates for Men+ are very similar between non-immigrants and immigrants, with a gap of less than one percentage point. For Women+, the difference is larger: non-immigrant women have an employment rate of approximately 92.3%, compared with approximately 88.9% for immigrant women.

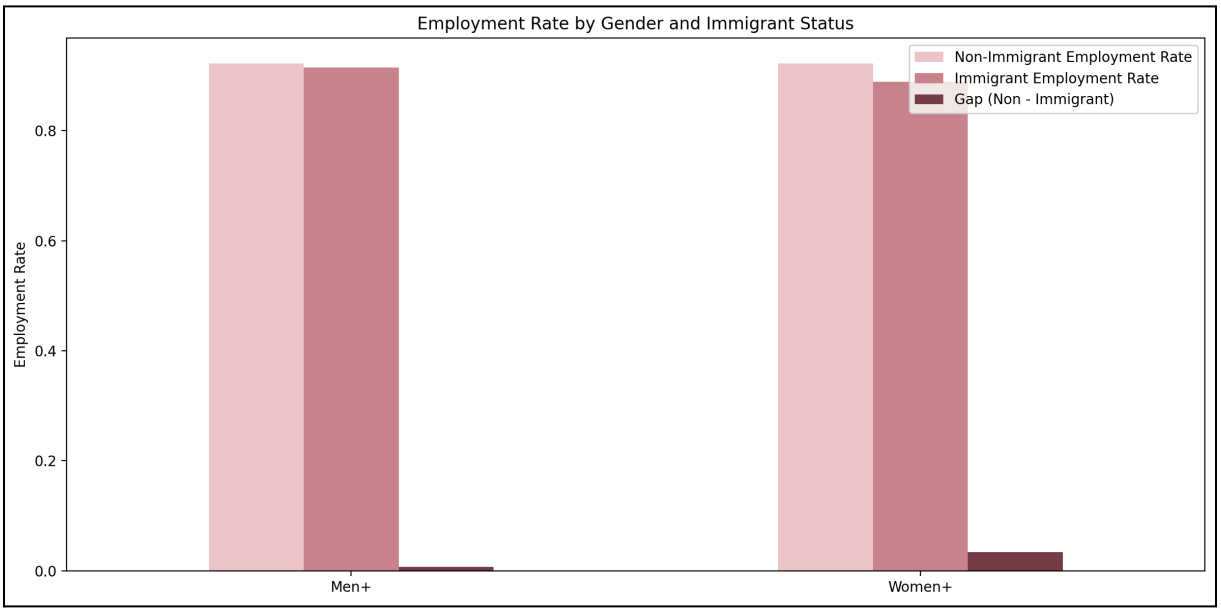

Figure 10: Employment rate by gender and immigrant status

Figure 11 highlights that the employment gap is not evenly distributed by gender. The gap is small for Men+ but more pronounced for Women+, suggesting that immigrant women may face intersecting labour market barriers related to gender, immigrant status, credential transferability, care responsibilities, or access to networks. Because the data are aggregated, this pattern should be interpreted as a group-level association rather than an individual-level explanation.

Figure 11: Employment rate gap by gender

Does Geography Matter?

Geography provides additional context for the employment gap. As shown in Figure 12, employment rates vary across provinces for both immigrant and non-immigrant groups. The direction of the gap is not uniform: immigrants have higher employment rates than non-immigrants in several Atlantic provinces, while non-immigrants have higher employment rates in provinces such as Quebec, Ontario, Alberta, and British Columbia.

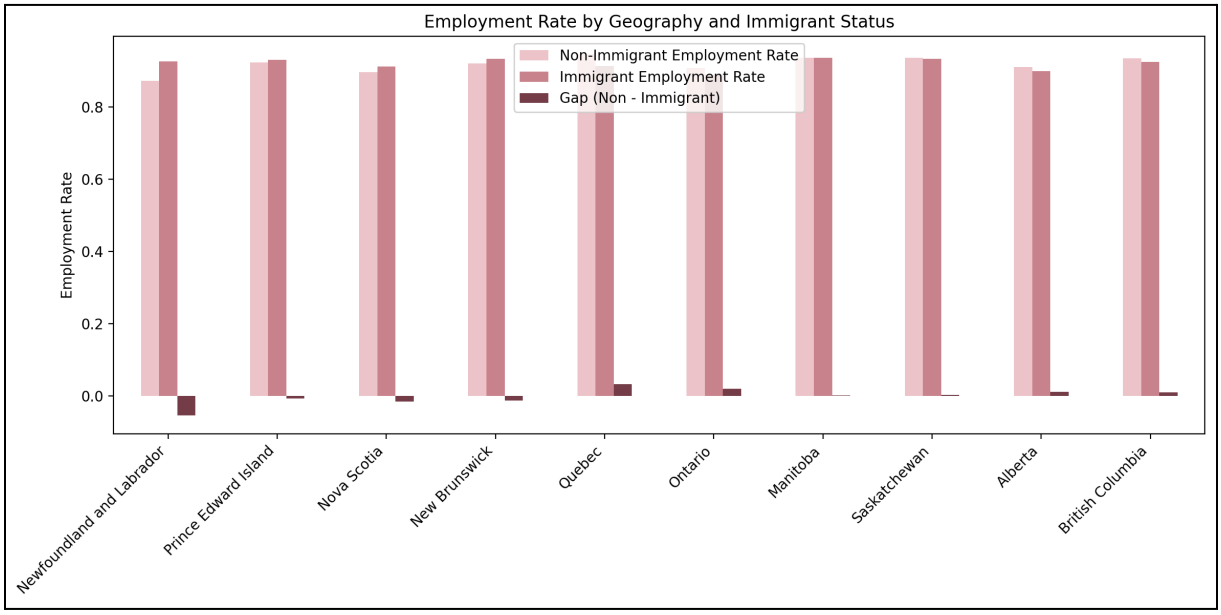

Figure 12: Employment rate by geography and immigrant status

As shown in Figure 13, the largest positive non-immigrant advantage appears in Quebec and Ontario, while the gap reverses in Newfoundland and Labrador, Nova Scotia, New Brunswick, and Prince Edward Island. This suggests that immigrant employment outcomes are shaped not only by individual characteristics such as education and age, but also by regional labour market conditions, settlement patterns, and provincial economic structures.

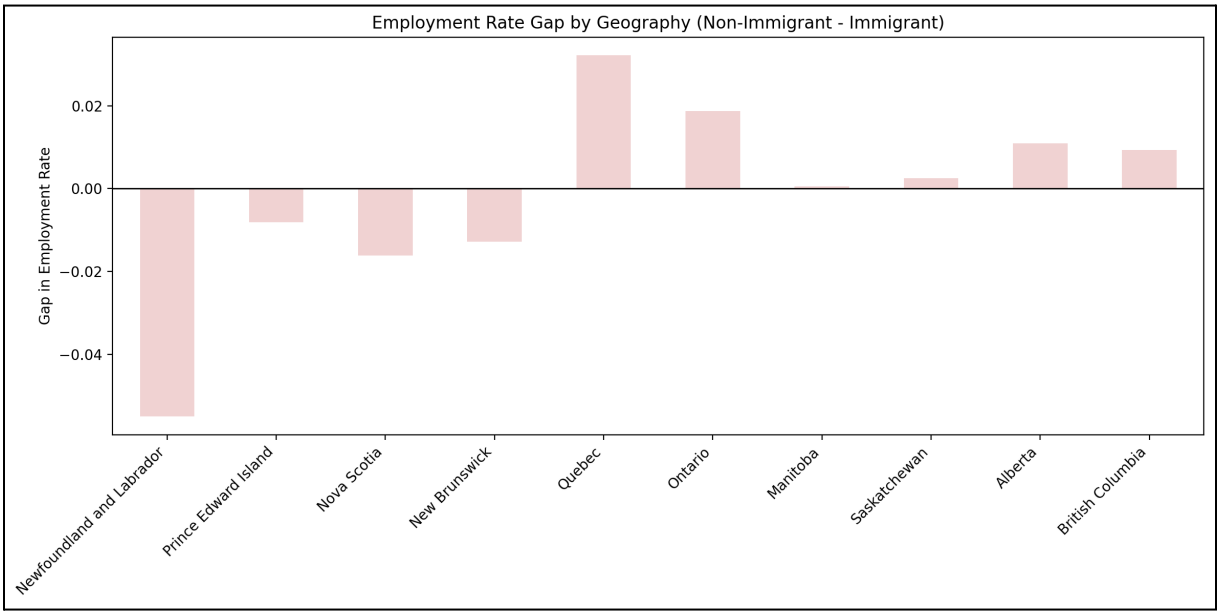

Figure 13: Employment rate gap by geography

As shown in Figure 13, the largest positive non-immigrant advantage appears in Quebec and Ontario, while the gap reverses in Newfoundland and Labrador, Nova Scotia, New Brunswick, and Prince Edward Island. This suggests that immigrant employment outcomes are shaped not only by individual characteristics such as education and age, but also by regional labour market conditions, settlement patterns, and provincial economic structures.

Feature Interpretation

The feature importance analysis suggests that the models were most sensitive to variables related to:

- field of study
- location of study
- age
- geography
- gender

In Logistic Regression, several field-of-study categories show strong coefficients, while Random Forest places greater emphasis on variables such as gender, location of study, and age. However, these findings should be interpreted with caution.

Given that both models demonstrated weak predictive performance, the feature importance results do not provide strong causal or explanatory evidence. Instead, they indicate that while observable characteristics are relevant, they do not capture the full complexity of employment outcomes.

Additional Interpretations of Subgroup Patterns

A more detailed examination of subgroup patterns suggests that the relationship between immigrant status and employment outcomes varies across age, education, gender, and geography in ways that reflect broader labour market dynamics.

Across age groups, the employment gap between immigrants and non-immigrants appears to be more pronounced among younger individuals and narrows in older age categories. One possible interpretation is that labour market entry may be more challenging for younger immigrants, particularly those who arrive as students or dependents. In these cases, limited Canadian work experience and language barriers may affect early employment outcomes.

At the same time, the narrowing gap among older age groups may reflect a gradual process of labour market integration. Over time, immigrants may accumulate local work experience, develop professional networks, and improve language proficiency, all of which can contribute to improved employment outcomes. However, this convergence does not necessarily imply equivalent long-term outcomes. Differences in job quality, income stability, access to benefits may still persist; potentially causing a lack of savings and the inability to retire.

Differences across education levels also reveal important patterns. The largest employment gaps appear in highly specialized fields, such as medicine, where international credentials may not be fully recognized or transferable to Canadian regulatory standards. This interpretation is consistent with existing research on credential recognition and occupational licensing barriers (ESDC, 2025).

In contrast, the smallest gaps are observed among individuals with little or no formal education. One possible explanation is that low-skilled occupations may rely less on formal credential recognition or be more accepting of language barriers, allowing immigrants and non-immigrants to compete more directly in these segments of the labour market. While this

may reduce observable employment differences, it does not necessarily imply equivalent job quality or long-term economic outcomes.

These patterns reinforce the broader conclusion of the study: employment outcomes are shaped not only by observable characteristics such as age, education, gender, and geography, but also by institutional and regional factors that influence how skills and experience are recognized and rewarded within the labour market.

Synthesis of Findings

Taken together, the modelling and descriptive results support three main conclusions:

1. Age, education, gender, and geography are important correlates of employment outcomes.
2. Observable demographic, education-related, and geographic variables do not strongly predict employment status.
3. Differences between immigrant and non-immigrant groups persist across comparable age, education, gender, and geographic categories, although the size and direction of the gap vary by subgroup.

This is a meaningful analytical result. It suggests that the remaining employment gap is likely influenced by factors not captured in the dataset, including language proficiency, credential recognition, labour market experience, institutional barriers, and regional labour market conditions.

Discussion of Research Impact and Practical Implications

The findings have both methodological and policy implications.

From a methodological perspective, the study demonstrates the value of combining predictive modelling with descriptive subgroup analysis. Model performance alone would suggest weak predictive power, but when paired with subgroup comparisons, a more informative conclusion emerges: observable characteristics are associated with employment outcomes but do not fully explain group differences.

From a policy perspective, the results suggest that improvements in formal education alone are unlikely to eliminate employment gaps. While immigrant labour market outcomes have improved in recent years (Hou, 2024), persistent differences indicate that structural barriers remain.

This has practical implications in several areas:

1. **Credential recognition:** Employment gaps among highly educated individuals suggest that the issue is not education itself, but how it is evaluated and utilized within the labour market.
2. **Early labour market integration:** Stronger differences among younger groups highlight the importance of transition supports, including internships, co-op opportunities, and employer connections.
3. **Settlement and language support:** Even though language is not directly modelled, prior research shows it remains an important determinant of labour market outcomes (Xu & Hou, 2023).
4. **Regional support:** Provincial variation suggests that immigrant employment integration is shaped partly by local labour market conditions, so policy responses may need to be tailored to regional employment structures and settlement contexts.

Overall, the findings support an interpretation of labour market integration as an institutional and structural process, rather than one determined solely by individual characteristics.

Limitations & Ethical Considerations

Limitations

This study is subject to several important limitations that shape how the results should be interpreted. Most notably, the dataset is aggregated rather than based on individual-level observations, the feature set omits several key variables, the modelling approach relies on sampling for computational feasibility, and the analysis is cross-sectional. Taken together, these constraints mean the findings are better understood as descriptive and predictive associations rather than causal evidence.

A key limitation lies in the structure of the dataset itself. Because each observation represents a grouped combination of characteristics rather than an individual, the analysis cannot capture variation within groups. This reduces the precision of the models and limits their ability to identify more nuanced relationships between characteristics and employment outcomes.

The feature set further restricts the analysis. While it includes demographic, educational, and geographic variables, it does not account for factors such as language proficiency, Canadian work experience, years since immigration, occupational licensing, industry, or social networks. Existing research highlights the importance of these variables in shaping immigrant labour market outcomes (Kaushik & Drolet, 2018; Xu & Hou, 2023; ESDC, 2025), so their absence likely contributes to the relatively weak predictive performance observed.

From a methodological standpoint, the use of stratified sampling was necessary to make the modelling process computationally feasible. However, this also means the models were not optimized for maximum predictive accuracy across the full dataset. Instead, the emphasis was on identifying broader patterns and limitations within a manageable and reproducible framework.

The cross-sectional nature of the analysis introduces an additional constraint. Since the study is based on data from the 2021 Census of Population, it does not capture changes over time or allow for causal inference. Although the findings are interpreted within a post-pandemic context, they do not directly reflect how labour market outcomes evolve.

Finally, the reliance on aggregated Census data as opposed to individual-level microdata, such as the Labour Force Survey (LFS), highlights a broader trade-off between coverage and analytical depth. While the Census offers comprehensive population coverage and detailed

subgroup breakdowns, LFS data would enable more granular modelling and time-series analysis. This distinction is important, as it helps explain some of the study's outcomes, particularly the limited predictive strength observed across models.

Causality Versus Association

The analysis identifies associations between observable characteristics and employment outcomes, but it does not establish causality. Because the data are cross-sectional and aggregated, the results cannot show that immigrant status, gender, age, education, or geography directly cause employment differences. The observed patterns may reflect unmeasured factors such as language proficiency, years since immigration, credential recognition, local labour demand, social networks, discrimination, or access to Canadian work experience. For this reason, the findings should be interpreted as group-level associations that point to possible structural mechanisms, not as causal estimates or individual-level predictions.

Ethical Considerations

This study also raises several important ethical considerations, particularly around how the results are interpreted and used. Many of the variables included, such as immigrant status, age, and gender, are socially constructed and closely tied to broader structural inequalities. Because of this, any patterns observed in the data should not be understood as reflecting inherent differences in employability between groups. Instead, they are better seen as signals of systemic factors that shape labour market outcomes.

The relatively weak predictive performance of the models adds another layer to this discussion. Since the models perform only marginally better than random classification, they are not suitable for real world decision making contexts such as hiring or candidate screening. Their value lies in highlighting patterns and limitations within the data, which reinforces that the findings are intended strictly for analytical and research purposes, not for individual level application.

There is also a risk of overgeneralization when working with aggregate data. The results capture trends at the group level, not individual experiences, so they need to be interpreted with care. Without thoughtful framing, there is a possibility of reinforcing stereotypes or attributing differences in employment outcomes to individual shortcomings rather than recognizing the structural and institutional factors at play.

Finally, the use of official statistical data brings a responsibility to interpret findings within a broader context. While the analysis identifies persistent differences in employment outcomes, these patterns are closely linked to factors such as credential recognition, barriers to labour

market entry, and institutional processes (Schimmele & Hou, 2024; ESDC, 2025). Keeping this context in mind helps ensure the results are used thoughtfully and do not unintentionally misrepresent the populations being studied.

Future Work & Recommendations

Future Research

There are several ways this study could be extended to deepen the analysis and address its current limitations. One natural next step would be to incorporate multiple Census periods. Including earlier cycles, such as 2017, along with future releases like the 2025 Census, would allow for a clearer view of long-term trends in immigrant labour market integration. This kind of approach could help determine whether the employment gaps observed here are consistent over time or shifting, and whether the COVID-19 period introduced lasting changes or simply reflected pre-existing structural patterns.

Another valuable extension would be to integrate individual-level data from the Labour Force Survey (LFS). While the Census offers strong population coverage and detailed subgroup breakdowns, LFS data would allow for more granular modelling and the inclusion of variables such as job tenure, occupation, industry, and hours worked. Bringing these sources together would support a more comprehensive analysis that captures both employment status and the underlying mechanisms driving labour market outcomes.

Building on this, future research could take a closer look at how immigrants and non-immigrants are distributed across occupations and industries. The current findings suggest that employment gaps persist even after accounting for factors like age, education, gender, and geography. This raises the possibility that differences in sectoral concentration may be playing a role, particularly if certain groups are more likely to work in industries with lower stability, income, or advancement opportunities. Exploring this dimension could provide important insight into the structural drivers of these patterns.

There is also room to better understand labour market entry pathways, especially for younger immigrants. The results indicate that employment gaps tend to be more pronounced in younger age groups, which points to the importance of early career experiences. Factors such as student status, transitions into the workforce, language proficiency, and access to Canadian work experience may all shape initial outcomes. Examining these elements in more detail could help explain how early disadvantages emerge and whether they persist or diminish over time.

Finally, incorporating variables related to credential recognition and skill transferability would add another important layer to the analysis. This is particularly relevant in regulated fields like healthcare and professional services, where barriers to recognizing international qualifications

can limit employment opportunities. The current results suggest that gaps vary across education levels, which may reflect these challenges. A more focused investigation could help quantify the extent to which credential recognition affects outcomes across different sectors.

Taken together, these extensions would move the analysis beyond identifying patterns and toward a more complete understanding of the mechanisms shaping immigrant labour market outcomes.

Practical Recommendations

The findings of this study suggest several practical implications for labour market policy and integration strategies.

1. **Strengthen credential recognition systems**
   Persistent employment gaps among highly educated individuals indicate that the issue is not solely educational attainment, but how credentials are evaluated and applied within the labour market. Improving the efficiency and transparency of credential recognition processes could help reduce these barriers.
2. **Support early labour market integration**
   The larger employment gaps observed among younger individuals suggest that labour market entry may be particularly challenging for immigrants. Expanding access to internships, co-op programs, and entry-level opportunities could help facilitate smoother transitions into the workforce.
3. **Enhance language and workplace readiness programs**
   Although language proficiency is not directly measured in this study, prior research indicates its importance. Continued investment in language training and workplace communication support may improve employment outcomes, particularly in early career stages.
4. **Improve access to Canadian work experience**
   Many employment barriers are linked to the requirement for local experience. Creating more structured pathways (mentorship programs, bridging programs, and employer partnerships) could help immigrants gain relevant experience more quickly.
5. **Encourage equitable evaluation of international education and experience**
   Employers play a key role in labour market integration. Encouraging more consistent and informed evaluation of international qualifications could help reduce disparities in employment outcomes.

Code & Reproducibility

This project includes a reproducible analytical workflow implemented in Python for data preprocessing, modelling, evaluation, and visualization. The analysis is based on aggregated data from **Statistics Canada Table 98-10-0442-01**, derived from the **2021 Census of Population**.

The reproducible workflow includes:

- data cleaning and preprocessing steps applied to the aggregated table
- a Jupyter Notebook used for model training, testing, and subgroup analysis
- exported result files summarizing model performance, cross-validation results, and feature importance
- visual outputs used to support the figures presented in the report

Repository Requirements

https://github.com/jayemm2210/TMU_CIND820

Suggested Reproducibility Statement

This project was designed with reproducibility in mind. All data preprocessing, modelling, evaluation, and visualization steps are fully documented in the accompanying Python notebook and repository files. The analysis is based on aggregated data from Statistics Canada Table 98-10-0442-01, derived from the 2021 Census of Population.

To ensure computational feasibility, stratified sampling was used during model training and cross-validation. These decisions are explicitly documented in the workflow and were applied consistently across all models to maintain comparability.

Because the dataset is aggregated rather than individual-level, the results reflect group-level patterns rather than individual predictions. Within these constraints, the provided code and documentation are sufficient to reproduce the analytical process, model comparisons, and descriptive findings presented in this study.

Research Questions

The findings suggest that observable demographic, education-related, and geographic characteristics offer only limited predictive power for employment outcomes. Both models produced accuracy and F1

scores close to 0.50, indicating that these variables capture some association with employment status, but not enough to support strong prediction on their own.

At the descriptive level, age, education, gender, and geography all help illuminate differences between immigrant and non-immigrant employment outcomes, but they do not fully account for them. Employment rates vary across these subgroups in meaningful ways, yet gaps remain across many comparable categories, and those gaps differ in both size and direction depending on the subgroup being considered.

The model comparison adds some nuance, though not a decisive distinction. Random Forest achieved a slightly stronger F1 score, while Logistic Regression offered greater interpretability and somewhat more stable performance. Still, because both models performed weakly overall, the more important takeaway is not which model “won,” but that the available variables were not sufficient to predict employment outcomes with much accuracy.

Taken together, the remaining differences point beyond the variables included in the dataset. They suggest that broader structural and institutional factors likely shape labour market outcomes in important ways. These may include credential recognition processes, language proficiency, access to Canadian work experience, labour market entry pathways, and regional labour market conditions, all of which may influence employment outcomes without being directly captured in the present analysis.

Conclusion

This study examined whether employment outcomes in Canada can be explained by observable demographic, education-related, and geographic characteristics, and whether differences between immigrant and non-immigrant groups persist when age, education, gender, and geography are taken into account.

Using aggregated data from the 2021 Census of Population, the analysis combined predictive modelling with descriptive subgroup comparisons. The modelling results showed that both Logistic Regression and Random Forest performed weakly, with accuracy close to 50% and F1 scores only slightly above 0.50. These results indicate that the selected variables, including age, education, gender, and geography, do not strongly predict employment status.

The descriptive analysis provides the study's most important substantive findings. Age, education, gender, and geography are associated with employment outcomes, but these associations should not be interpreted as causal effects. These variables do not fully explain the observed differences between immigrant and non-immigrant groups. While employment rates generally improve with higher education and vary across age groups, gender categories, and provinces, a gap remains across many comparable categories.

Further examination of subgroup patterns suggests that this gap is not uniform. Differences are more pronounced among younger individuals and tend to narrow in older age groups, which may reflect gradual labour market integration over time. Similarly, variation across education levels indicates that employment gaps are larger in highly regulated or specialized fields, where credential recognition may play a role, and smaller in lower-skilled categories where formal qualifications are less central.

Taken together, these findings suggest that employment outcomes are shaped not only by observable characteristics, but also by structural, institutional, and regional factors that influence how skills and experience are recognized within the labour market. This interpretation is consistent with existing research emphasizing the importance of credential recognition, language proficiency, access to Canadian work experience, and local labour market conditions.

The remaining gaps suggest that structural and institutional barriers not captured in the dataset are central to understanding immigrant labour market integration.

Overall, the study demonstrates that while demographic, education-related, and geographic variables are important, they are not sufficient to explain employment outcomes or eliminate group-level differences. Future research should therefore focus on incorporating more detailed individual-level data and exploring the mechanisms underlying labour market integration, particularly in relation to institutional barriers, regional labour market conditions, and labour market entry pathways.

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